

Lab 3

Joseph Leone

4-25-2026

CDA 3203 Computer Logic Design

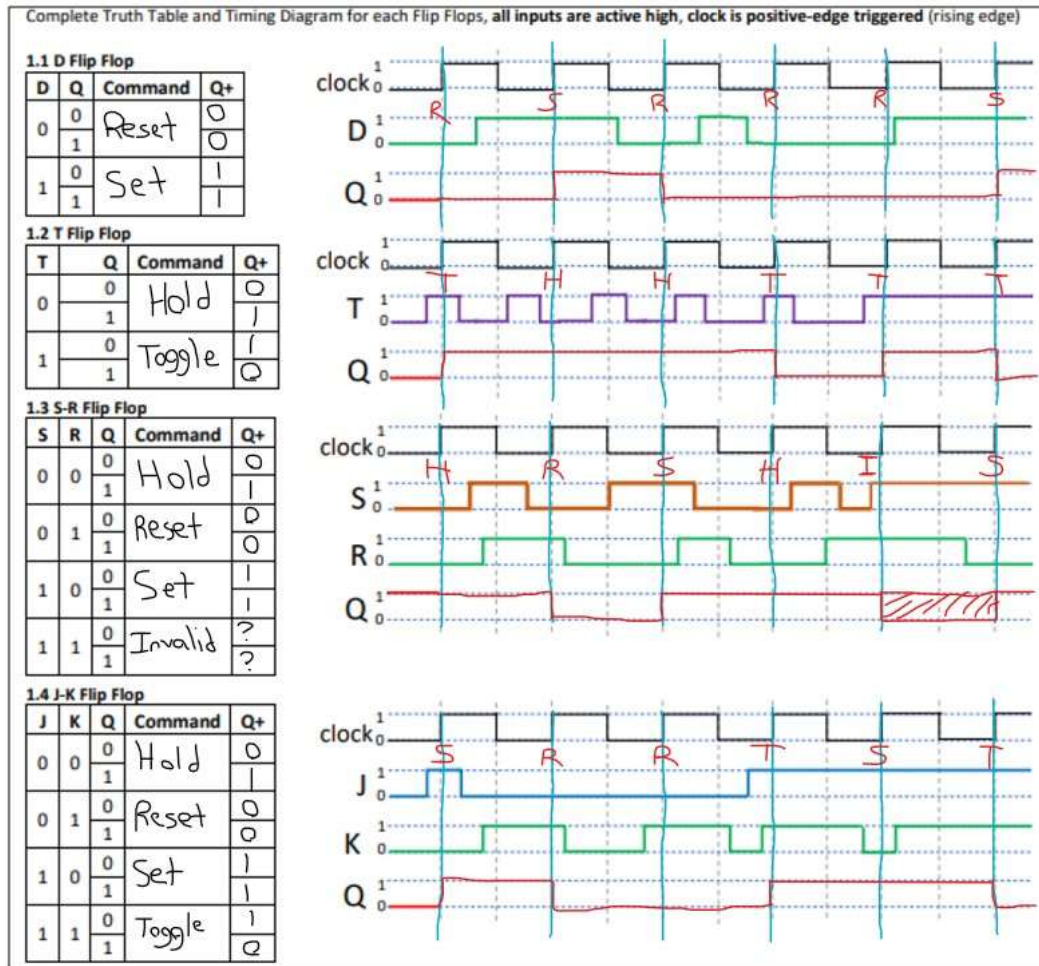
Spring 2026

Dr. Maria Petrie

Florida Atlantic University

3.0) Handwork (40 pts)_____

Fill in the Diagrams and Truth Tables below, you can delete the image below and paste yours in.
Or if you can draw/annotate on top of the image below – using OneNote or another notetaking app.



3.1) Simulating a Data Flip Flop (DFF)(20pts)

Dr. Petrie gives you the code for this Entity. Watch the videos!

New Project Wizard: Summary [page 5 of 5]

When you click Finish, the project will be created with the following settings:

Project directory:
G:/My Drive/Documents/Study/CDA3203/labs/Lab3_JosephLeone/

Project name: DFF_JosephLeone
Top-level design entity: DFF_JosephLeone
Number of files added: 0
Number of user libraries added: 0

Device assignments:
Family name: Stratix II
Device: AUTO

EDA tools:
Design entry/synthesis: <None>
Simulation: <None>
Timing analysis: <None>

Operating conditions:
Core voltage: n/a
Junction temperature range: n/a

< Back Next > Finish Cancel

abc DFF_JosephLeone.vhd Compiler 1

```
1  -- Data Flip Flop
2  -- CDA 3203-042
3  -- Joseph Leone
4
5  Library ieee;
6  USE IEEE.std_logic_1164.ALL;
7
8  ENTITY DFF_JosephLeone IS
9  PORT (D, Clock : IN STD_LOGIC;
10         Q : INOUT STD_LOGIC);
11  END DFF_JosephLeone;
12
13  Architecture Behavior OF DFF_JosephLeone IS
14  BEGIN
15  PROCESS (Clock)
16  BEGIN
17  IF (rising_edge(Clock)) THEN
18  Q <= D;
19  END IF;
20  END PROCESS;
21  END Behavior;
```

DFF_JosephLeone.vhd

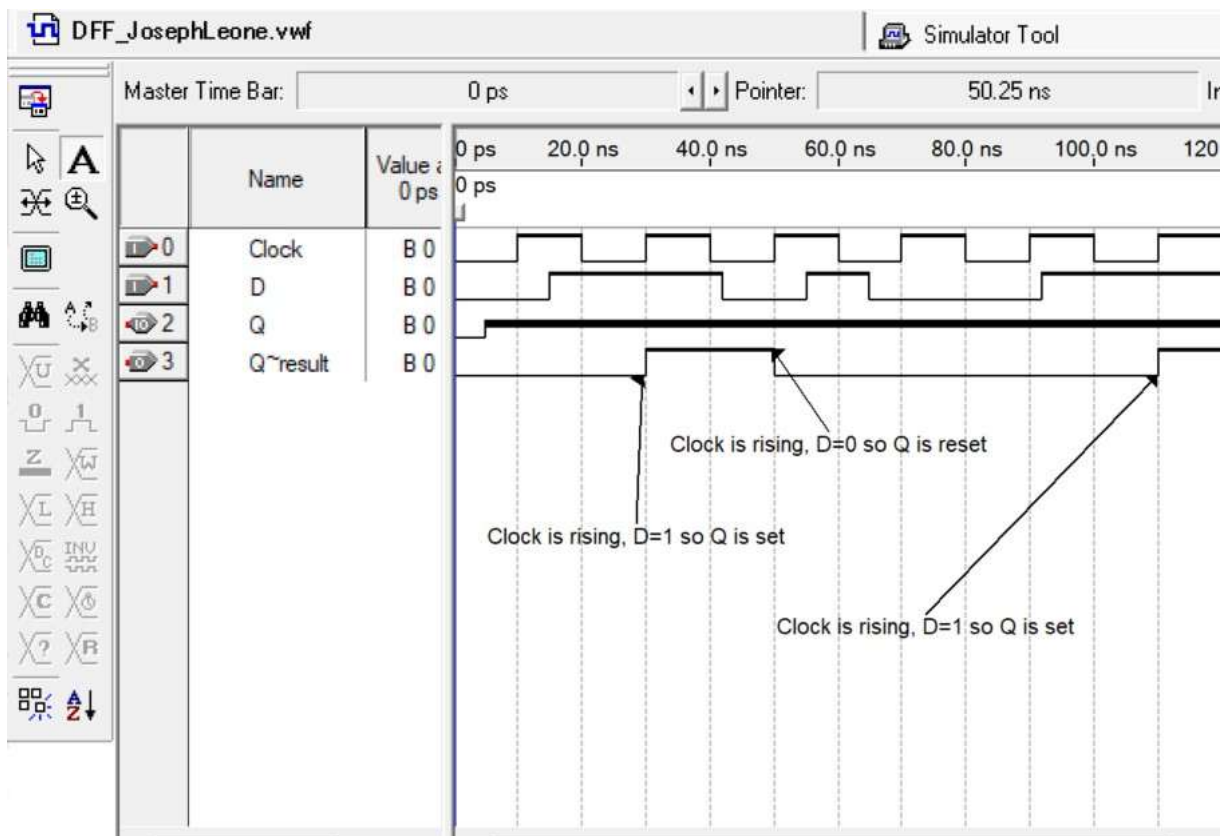
Compiler Tool

Quartus II

Full Compilation was successful (11 warnings)

OK

```
1  -- Data Flip Flop
2  -- CDA 3203-042
3  -- Joseph Leone
4
5  Library ieee;
6  USE IEEE.std_logic_1164.ALL;
7
8  ENTITY DFF_JosephLeone IS
9  PORT (D, Clock : IN STD_LOGIC;
10       Q : INOUT STD_LOGIC);
11  END DFF_JosephLeone;
12
13  Architecture Behavior OF DFF_JosephLeone IS
14  BEGIN
15  PROCESS (Clock)
16  BEGIN
17      IF (rising_edge(Clock)) THEN
18          Q <= D;
19      END IF;
20  END PROCESS;
21  END Behavior;
```



3.2) Designing and simulating a 1-Bit Register (40pts) _____

New Project Wizard: Summary [page 5 of 5]

When you click Finish, the project will be created with the following settings:

Project directory:
G:/My Drive/Documents/study/cda3203/labs/Lab3_JosephLeone/

Project name: 1BIT_JosephLeone

Top-level design entity: 1BIT_JosephLeone

Number of files added: 0

Number of user libraries added: 0

Device assignments:

Family name: Stratix II

Device: AUTO

EDA tools:

Design entry/synthesis: <None>

Simulation: <None>

Timing analysis: <None>

Operating conditions:

Core voltage: n/a

Junction temperature range: n/a

< Back Next > Finish Cancel

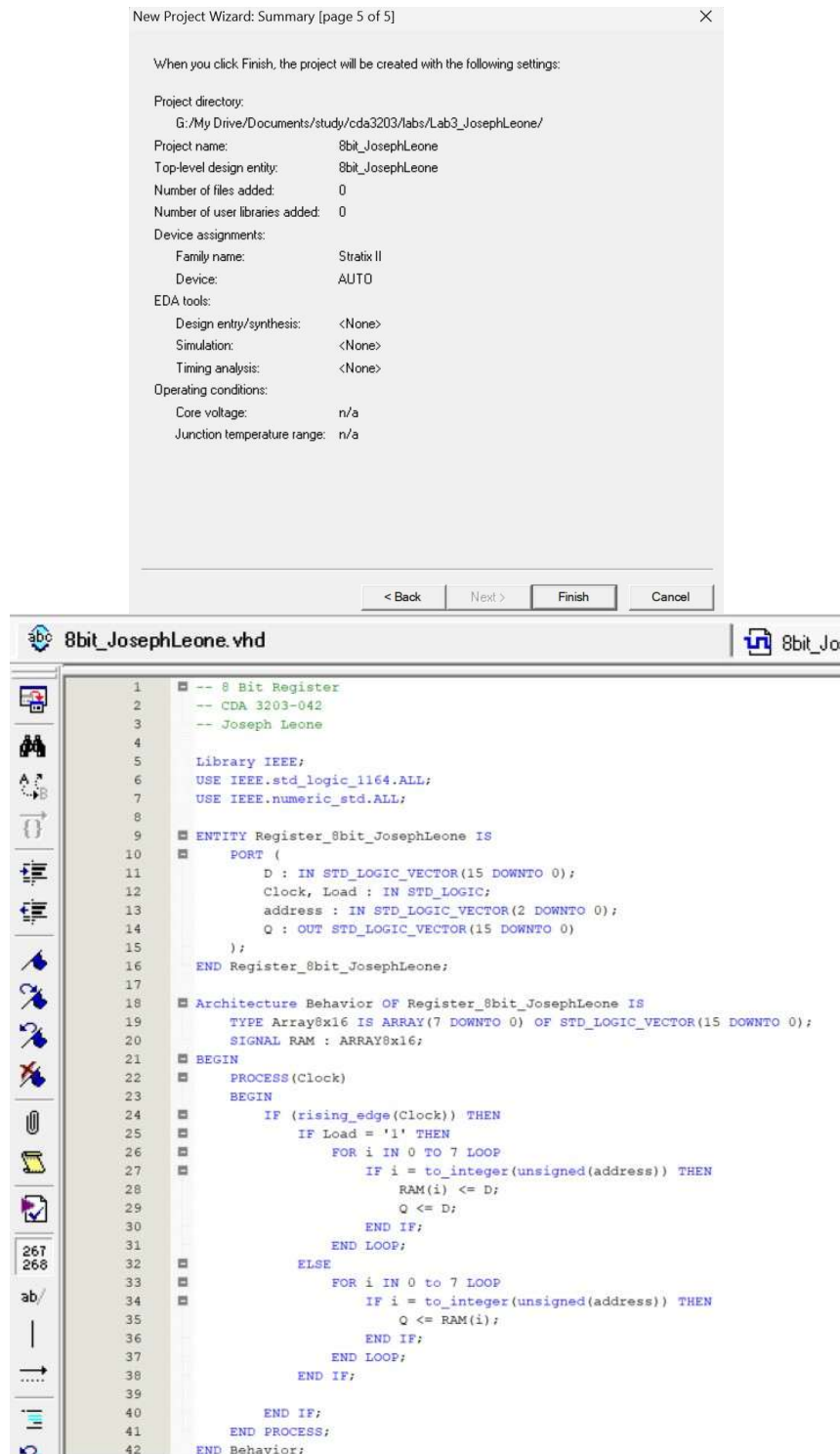
1BIT_JosephLeone.vhd 1BIT_JosephLeone.vwf

```
1  -- 1 Bit Register
2  -- CDA 3203-042
3  -- Joseph Leone
4
5  Library IEEE;
6  USE IEEE.std_logic_1164.ALL;
7
8  ENTITY Register_1bit_JosephLeone IS
9  PORT (D, Clock, Load : IN STD_LOGIC;
10       Q : INOUT STD_LOGIC);
11  END Register_1bit_JosephLeone;
12
13  Architecture Behavior OF Register_1bit_JosephLeone IS
14  BEGIN
15  PROCESS (Clock)
16  BEGIN
17      IF (rising_edge(Clock)) THEN
18          IF Load = '1' THEN
19              Q <= D;
20          END IF;
21      END IF;
22  END PROCESS;
23  END Behavior;
```


3.3) EXTRA CREDIT ENTITY

Designing and Simulating a 8-Bit Register (20pts)

No Structural code, ONLY behavioral. This is an ACTIVE HIGH entity. You are expected to ANNOTATE the. vwf behavior and why your timing diagram is correct for full credit.



The image shows two windows from a VHDL development environment. The top window is the 'New Project Wizard: Summary' dialog, page 5 of 5. It displays the following settings:

- Project directory: G:/My Drive/Documents/study/cda3203/labs/Lab3_JosephLeone/
- Project name: 8bit_JosephLeone
- Top-level design entity: 8bit_JosephLeone
- Number of files added: 0
- Number of user libraries added: 0
- Device assignments:
 - Family name: Stratix II
 - Device: AUTO
- EDA tools:
 - Design entry/synthesis: <None>
 - Simulation: <None>
 - Timing analysis: <None>
- Operating conditions:
 - Core voltage: n/a
 - Junction temperature range: n/a

The bottom window shows the VHDL code for '8bit_JosephLeone.vhd'. The code is as follows:

```
1  -- 8 Bit Register
2  -- CDA 3203-042
3  -- Joseph Leone
4
5  Library IEEE;
6  USE IEEE.std_logic_1164.ALL;
7  USE IEEE.numeric_std.ALL;
8
9  ENTITY Register_8bit_JosephLeone IS
10     PORT (
11         D : IN STD_LOGIC_VECTOR(15 DOWNTO 0);
12         Clock, Load : IN STD_LOGIC;
13         address : IN STD_LOGIC_VECTOR(2 DOWNTO 0);
14         Q : OUT STD_LOGIC_VECTOR(15 DOWNTO 0)
15     );
16 END Register_8bit_JosephLeone;
17
18 ARCHITECTURE Behavior OF Register_8bit_JosephLeone IS
19     TYPE Array8x16 IS ARRAY(7 DOWNTO 0) OF STD_LOGIC_VECTOR(15 DOWNTO 0);
20     SIGNAL RAM : Array8x16;
21 BEGIN
22     PROCESS(Clock)
23     BEGIN
24         IF (rising_edge(Clock)) THEN
25             IF Load = '1' THEN
26                 FOR i IN 0 TO 7 LOOP
27                     IF i = to_integer(unsigned(address)) THEN
28                         RAM(i) <= D;
29                         Q <= D;
30                     END IF;
31                 END LOOP;
32             ELSE
33                 FOR i IN 0 TO 7 LOOP
34                     IF i = to_integer(unsigned(address)) THEN
35                         Q <= RAM(i);
36                     END IF;
37                 END LOOP;
38             END IF;
39         END IF;
40     END PROCESS;
41 END Behavior;
```

8bit_JosephLeone.vhd

8bit_JosephLeone.vwf

1

-- 8 Bit Register

2

-- CDA 3203-042

3

-- Joseph Leone

4

Library IEEE;

5

USE IEEE.std_logic_1164.ALL;

6

USE IEEE.numeric_std.ALL;

7

8

9

ENTITY Register_8bit_JosephLeone IS

10

PORT (

11

D : IN STD_LOGIC_VECTOR(15 DOWNT0 0);

12

Clock, Load : IN STD_LOGIC;

13

address : IN STD_LOGIC_VECTOR(2 DOWNT0 0);

14

Q : OUT STD_LOGIC_VECTOR(15 DOWNT0 0)

15

);

16

END Register_8bit_JosephLeone;

17

18

Architecture Behavior OF Register_8bit_JosephLeone IS

19

TYPE Array8x16 IS ARRAY(7 DOWNT0 0) OF STD_LOGIC_VECTOR(15 DOWNT0 0);

20

SIGNAL RAM : ARRAY8x16;

21

BEGIN

22

PROCESS(Clock)

23

BEGIN

24

IF (rising_edge(Clock)) THEN

25

IF Load = '1' THEN

26

FOR i IN 0 TO 7 LOOP

27

IF i = to_integer(unsigned(address)) THEN

28

RAM(i) <= D;

29

Q <= D;

30

END IF;

31

END LOOP;

32

ELSE

33

FOR i IN 0 TO 7 LOOP

34

IF i = to_integer(unsigned(address)) THEN

35

Q <= RAM(i);

36

END IF;

37

END LOOP;

38

END IF;

39

END IF;

40

END PROCESS;

41

END Behavior;

Quartus II

Full Compilation was successful (6 warnings)

OK

